

# GPS — Teri Location Kaise Pata Chalti Hai?

**GPS = 24–31 satellites, ~20,000 km upar**

India = NavIC

# Signal me: satellite ki location + bhejne ka time

One-way!

$$\text{Distance} = \text{Speed} \times \text{Time}$$

Light speed ~3 lakh km/s

**Ek distance = ek sphere, ek pakka point  
nahi**



**Trilateration — teen circles ek point pe  
milte hain**

$$\underline{3 = 2D}$$

4th = altitude + clock fix

# **Relativity correction**

Warna roz kai km galat!



**Multipath, ionosphere, dewaarein =  
error**

**Assisted GPS = GPS + Wi-Fi + towers +  
sensors**

**Maps, delivery, kheti, rescue, banking —  
sab GPS pe**

**Quick Brain Check! Video PAUSE karo —  
3 sawaal**

**Answers: Speed × Time**

Trilateration (4 sats)



## **Next: Why Is the Ocean Salty?**

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